MATH 152 **QUIZ 8** NAME Solutions

Mrs. Bonny Tighe 25 points

11.8 TA_____ Wed 11/10/2021

1. Find the radius of convergence and interval of convergence of the power series.

4pts a) $\sum_{n=1}^{\infty} \frac{(-1)^n\, 5^n\, (x-2)^n}{1\cdot 3\cdot 5\cdots(2n-1)}$

$$\lim_{n\to\infty} \left| \frac{5\cdot 5^n (x-2)(x-2)^n}{(1\cdot 3\cdot 5\cdots(2n-1)(2n+1))} \cdot \frac{(1\cdot 3\cdot 5\cdots(2n-1))}{5^n (x-2)^n} \right|$$

$$\lim_{n\to\infty} \left| 5(x-2)\frac{1}{2n+1} \right| = 0 < 1 \text{ always}$$

$R = \infty$ $I = (-\infty, \infty)$

4pts b) $\sum_{k=0}^{\infty} \frac{(-1)^k (x+2)^{2k}}{9^k}$ Can use Geometric Series Test. Don't need to test endpoints

$$\sum_{k=0}^{\infty} (-1)^k \frac{(x+2)^{2k}}{3^{2k}} = \sum_{k=0}^{\infty} (-1)^k \left(\frac{(x+2)^2}{3^2}\right)^k = \sum_{k=0}^{\infty} \left(\frac{-(x+2)^2}{3^2}\right)^k$$

$|r| = \left|\frac{-(x+2)^2}{3^2}\right| < 1$ $|(x+2)^2| < 3^2$ $|x+2| < 3$

$-3 < x+2 < 3$

$-5 < x < 1$

$R = 3$

$I = (-5, 1)$

2. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x+1)^{2n+1}}{3^{n+1}(n+3)}$ as a polynomial to the fifth degree.

4pts

$$\frac{x+1}{3(3)} - \frac{(x+1)^3}{3^2(4)} + \frac{(x+1)^5}{3^3(5)}$$

$n=0$ $n=1$ $n=2$

3. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(x-2)^n}{\sqrt{n+5}(3^n)}$

6pts

$\lim_{n\to\infty} \left| \frac{(x-2)(x-2)^n}{3\cdot 3^n\sqrt{n+6}} \cdot \frac{\sqrt{n+5}\,3^n}{(x-2)^n} \right| = \lim_{n\to\infty} \left| \frac{(x-2)}{3}\sqrt{\frac{n+5}{n+6}} \right| = \left|\frac{x-2}{3}\right| < 1$

$|x-2| < 3 \qquad -3 < x-2 < 3 \qquad -1 < x < 5$

$R = 3$

$I = [-1, 5)$

Test Endpoints

$x = -1$

$\sum_{n=1}^{\infty} \frac{(-3)^n}{3^n(\sqrt{n+5})} = \sum_{n=1}^{\infty} \frac{(-1)^n 3^n}{3^n\sqrt{n+5}} = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+5}}$ Alternating Series Test conditionally convergent

$x = 5$

$\sum_{n=1}^{\infty} \frac{3^n}{3^n\sqrt{n+5}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n+5}}$ Divergent Limit Comparison Test

b) $\sum_{n=1}^{\infty} \frac{(2\cdot 5\cdot 8\cdots(3n-1))(x-3)^n}{7^n}$

4pts

$\lim_{n\to\infty} \left| \frac{(2\cdot 5\cdot 8\cdots(3n-1)(3n+2))(x-3)(x-3)^n}{7\cdot 7^n} \cdot \frac{7^n}{(2\cdot 5\cdot 8\cdots(3n-1))(x-3)^n} \right|$

$\lim_{n\to\infty} \left| (3n+2)\left(\frac{x-3}{7}\right) \right| = \infty < 1$ never

$R = 0$

$I = \{3\}$

3pts

4. If $\sum_{n=0}^{\infty} c_n x^n$ is convergent when $x = -4$ and divergent when $x = 7$, decide which of the following are convergent or divergent. a) $\sum_{n=0}^{\infty} c_n(-3)^n$ Convergent

b) $\sum_{n=0}^{\infty} c_n$ Convergent c) $\sum_{n=0}^{\infty} c_n 9^n$ Divergent

MATH 152 **QUIZ 8A** NAME Solution

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1. Find the radius of convergence and interval of convergence of the power series.

4pts a) $\sum_{n=1}^{\infty} \frac{(1\cdot 3\cdot 5\cdots(2n-1))(3x-1)^n}{7^{n+1}}$

$\lim_{n\to\infty} \left| \frac{1\cdot 3\cdot 5\cdots(2n-1)(2n+1)(3x-1)(3x-1)^n}{7\cdot 7^{n+1}} \cdot \frac{7^{n+1}}{(1\cdot 3\cdot 5\cdots(2n-1))(3x-1)^n} \right|$

$\lim_{n\to\infty} \left| 2n+1 \left(\frac{3x-1}{7}\right) \right| = \infty < 1$ never so

$R = 0$

$I = \{1/3\}$

4pts b) $\sum_{k=0}^{\infty} \frac{(x-3)^{2k}}{4^k}$ Can use Geometric Series test so don't need to test the endpoints

$\sum_{k=0}^{\infty} \frac{(x-3)^{2k}}{2^{2k}} = \sum_{k=0}^{\infty} \left(\frac{(x-3)^2}{2^2}\right)^k$ $|r| = \left|\frac{(x-3)^2}{2^2}\right| < 1$

$|(x-3)^2| < 4$

$|x-3| < 2$

$-2 < x-3 < 2$

$1 < x < 5$

$R = 2$

$I = (1, 5)$

2. Expand $\sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{3^{n+1}(n+4)}$ as a polynomial to the fifth degree.

4pts

$\frac{(x-1)}{3(4)} - \frac{(x-1)^3}{9(5)} + \frac{(x-1)^5}{27(6)}$

$n=0$ $n=1$ $n=2$

6 pts

3. Find the radius of convergence and interval of convergence of the power series.

a) $\sum_{n=1}^{\infty} \frac{(x+1)^n}{\sqrt{n+2}(4^n)}$

$\lim_{n\to\infty} \left| \frac{(x+1)(x+1)^n}{\sqrt{n+3}\ 4\cdot 4^n} \cdot \frac{4^n\sqrt{n+2}}{(x+1)^n} \right| = \lim_{n\to\infty} \left| \frac{x+1}{4} \cdot \sqrt{\frac{n+2}{n+3}} \right| = \left|\frac{x+1}{4}\right| < 1$

$R = 4$

$I = [-5, 3)$

Test the end points

$|x+1| < 4$

$-4 < x+1 < 4$

$-5 < x < 3$

$x=-5$ $\sum_{n=1}^{\infty} \frac{(-4)^n}{\sqrt{n+2}\,4^n} = \sum_{n=1}^{\infty} \frac{(-1)^n 4^n}{\sqrt{n+2}\,4^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+2}}$ Conditionally Convergent by the Alternating Series Test

$x=3$ $\sum_{n=1}^{\infty} \frac{4^n}{\sqrt{n+2}\,4^n} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n+2}}$ Divergent, Limit Comparison Test

4 pts

b) $\sum_{n=1}^{\infty} \frac{(-1)^n\ 6^n\ (x-1)^n}{1\cdot 4\cdot 7\cdots(3n-2)}$

$\lim_{n\to\infty} \left| \frac{6\cdot 6^n\ (x-1)(x-1)^n}{(1\cdot 4\cdot 7\cdots(3n-2)(3n+1))} \cdot \frac{(1\cdot 4\cdot 7\cdots(3n-2))}{6^n\ (x-1)^n} \right|$

$\lim_{n\to\infty} \left| 6(x-1)\left(\frac{1}{3n+1}\right)\right| = 0 < 1$ always

$R = \infty$

$I = (-\infty, \infty)$

3 pts

4. If $\sum_{n=0}^{\infty} c_n x^n$ is convergent when $x = -4$ and divergent when $x = 7$, decide which of the following are convergent or divergent. a) $\sum_{n=0}^{\infty} c_n(-3)^n$ Convergent

b) $\sum_{n=0}^{\infty} c_n$ Convergent c) $\sum_{n=0}^{\infty} c_n 9^n$ Divergent